EXPERT INTELLIGENCE ASSISTANT

ProcureIQ

AI-powered procurement intelligence
from business request to confident decision

AI RESEARCH | VENDOR INTELLIGENCE | HUMAN APPROVAL



Hackathon Project | Proposed solution architecture

The procurement problem

Purchasing decisions often require research across fragmented sources and teams.

Fragmented supplier information

Prices, specifications, warranty and delivery details are scattered across websites, PDFs and emails.

Manual comparison work

Teams spend time collecting, normalizing and checking vendor information before they can compare options.

Limited decision traceability

Recommendations may lack clear evidence, consistent criteria or a record of who approved the next step.

The opportunity

Turn a purchasing request into a structured, evidence-backed decision — while keeping people in control of consequential actions.

Introducing ProcureIQ

A specialized procurement solution inside Expert Intelligence Assistant.

One connected procurement workflow

- Capture purchasing needs and constraints.
- Use AI to structure requirements and identify gaps.
- Research suppliers and collect source-backed evidence.
- Compare vendors using configurable criteria.
- Prepare recommendations for expert review.
- Manage approvals, RFQs and order tracking.



From request to recommendation — with an auditable human decision.

Who uses ProcureIQ?

Designed for collaboration between business teams, experts and approvers.

Customer / Department

Creates a purchase request, explains needs, answers clarification questions and reviews recommendations.

REQUEST OWNER

AI / Transformation Expert

Scopes the problem, runs research, evaluates evidence and prepares a recommendation.

WORKFLOW LEAD

Procurement Approver

Reviews the proposed vendor and authorizes RFQs or purchase actions within policy.

DECISION CONTROL

The AI supports research and analysis; authorized people retain control over final decisions.

Core product modules

A modular workspace covering the procurement lifecycle.

Purchase Requests

Capture products, quantities, budgets, specifications and delivery constraints.

Vendor Discovery

Find relevant suppliers and collect publicly available information.

Comparisons & Scoring

Normalize evidence and evaluate vendors against configurable criteria.

RFQs & Quotations

Prepare RFQ drafts and record vendor quotation responses.

Approvals

Route recommendations and purchasing actions to authorized reviewers.

Purchase Orders & Reports

Track order status and analyze procurement performance.

Each module shares the same customer, engagement, user and audit context from EIA.

How the AI intelligence layer works

Combine language understanding, web research and deterministic evaluation.

LLM — Requirement Understanding	Extract specifications, identify missing details and generate clarification questions.
CHRONICLE — Research Intelligence	Plan research, discover suppliers, investigate options, validate findings and synthesize results.
Browser Use — Web Evidence Collection	Visit supplier pages and collect public product, price, warranty and delivery information.
ProcureIQ — Procurement Logic	Normalize data, calculate weighted scores, manage workflow states and generate reports.

Important: verify provider compatibility and actual integrations before claiming a capability is operational.

End-to-end workflow

Example: an IT department needs 50 business laptops.

1 **Submit request**

50 laptops + specs

2 **Clarify needs**

Budget, deadline, warranty

3 **Research vendors**

CHRONICLE + Browser Use

4 **Validate evidence**

Sources + missing data

5 **Compare options**

Price, fit, delivery

6 **Recommend**

Explain trade-offs

7 **Approve**

Human decision

8 **Track order**

RFQ / PO / delivery

RFQ sending, purchase submission and other external actions require explicit authorization.

Evidence-based vendor evaluation

A transparent comparison helps experts understand both fit and uncertainty.

Evaluation	Vendor A	Vendor B	Ve
Specification fit	Strong match	Strong match	P
Quoted / listed price	$700 / unit	$750 / unit	$
Warranty	2 years	3 years	1
Delivery estimate	15 days	20 days	3
Evidence confidence	High	Medium	M

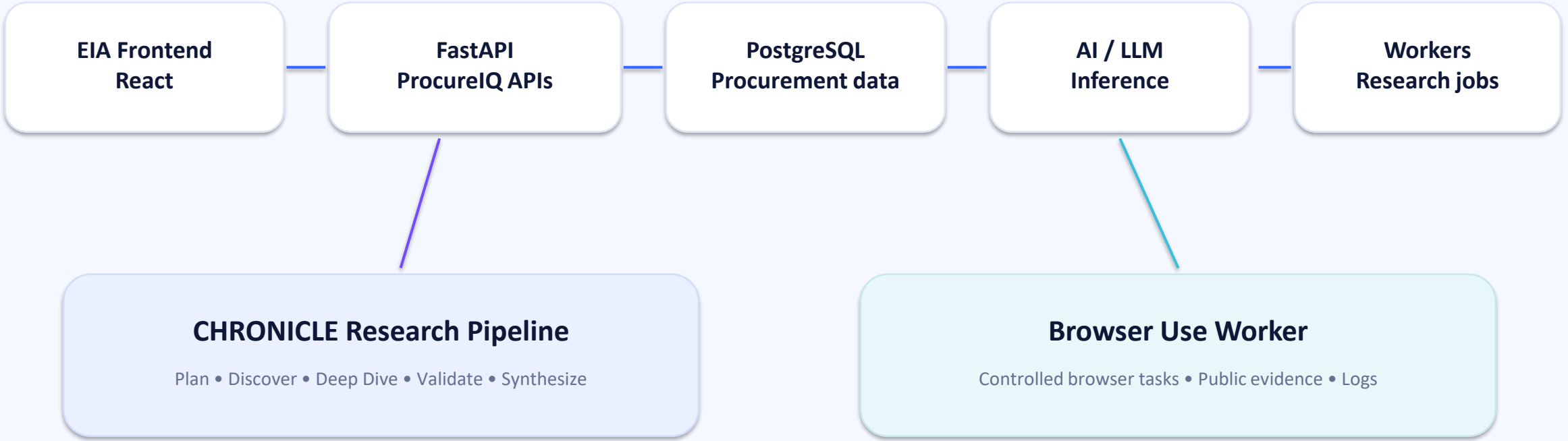
Illustrative scoring model

Specification fit	30%
Price	25%
Delivery	20%
Warranty & support	15%
Evidence confidence	10%

Scores are calculated by backend logic using configurable weights. Missing or unverified information is flagged — not silently treated as fact.

Proposed technical architecture

ProcureIQ is a specialized module within EIA, reusing shared platform services.



Deployment target: Vultr VM for app/backend and workers; PostgreSQL and Vultr Serverless Inference subject to architecture and compatibility validation.

Trust, security and human control

Procurement workflows need traceability, permissions and safe automation boundaries.

Human approval

Require authorization before sending RFQs, submitting orders or making commitments.

Evidence traceability

Store source URLs, timestamps and extracted evidence with research findings.

Untrusted web content

Treat supplier pages as data, not instructions; guard against prompt injection.

Isolation & secrets

Run browser tasks in controlled sessions; keep credentials outside browser tasks.

Role-based access

Enforce customer, expert and approver permissions in backend APIs.

Audit trail

Record important changes, approvals and procurement decisions.

MVP scope and delivery plan

Prioritize one complete, demonstrable procurement journey before expanding features.

01 Core workflow	Purchase request form • Request management • Vendor records
02 Research	LLM requirement analysis • CHRONICLE • Browser Use evidence collection
03 Intelligence	Vendor comparison • Configurable scoring • Recommendations
04 Governance	Human approvals • RFQ drafts • Quotation records
05 Deployment	Tests • Security checks • Vultr deployment • Demo preparation

MVP demo scenario: 50 laptops → research 3 suppliers → compare evidence → recommend → approve.

PROCUREIQ

Better procurement decisions, backed by evidence.

ProcureIQ connects business requirements, AI research and human judgment in one procurement workflow.

Research

Find and validate supplier information.

Compare

Understand trade-offs with transparent criteria.

Approve

Keep people in control of decisions.

Questions & discussion